Week 3 Level 1: Introduction

When This Happens...

Events start code; conditionals choose what happens.

What you will learn

- How **events** (like a key press or a click) start your scripts.
- How **if** and **if-else** let a sprite choose what to do.
- Why an **if** needs a **forever** loop to keep checking.
- How to give kind feedback: **two stars and a wish**.

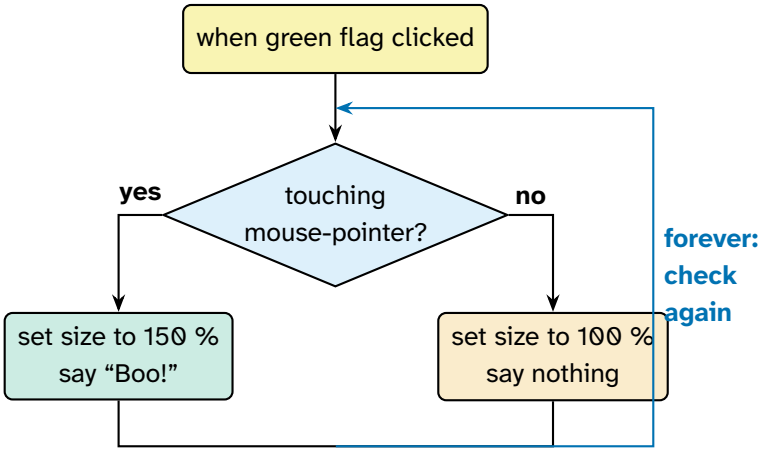
The big idea

Events are doorbells. A script that starts with a yellow **Events** hat block waits quietly. When its doorbell rings (the flag is clicked, a key is pressed, the sprite is clicked), it runs.



Conditionals ask a yes/no question. An **if** block runs the blocks inside it only when the answer is “yes”. An **if-else** block has two mouths: “yes” blocks and “no” blocks. Exactly one of them runs.

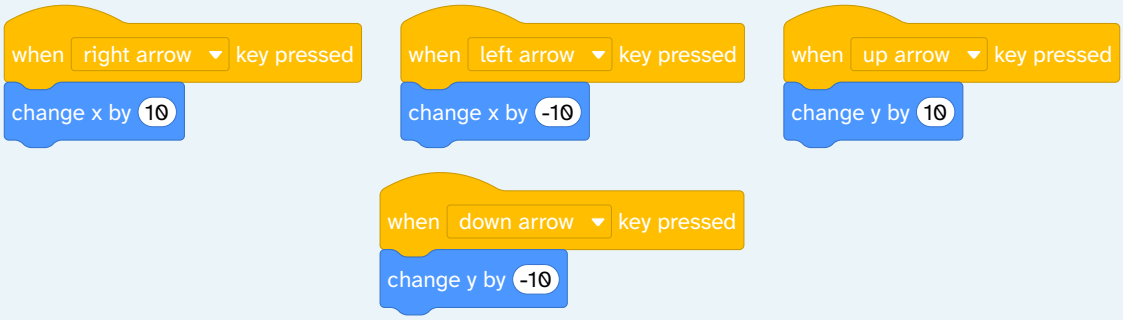
Keep checking! An **if** checks only once, when it runs. To keep watching, put it inside **forever**, like a goalkeeper who never stops watching the ball.



Exercise 3.1: Arrow-Key Explorer

Make a sprite move with the arrow keys. Each key is its own doorbell.

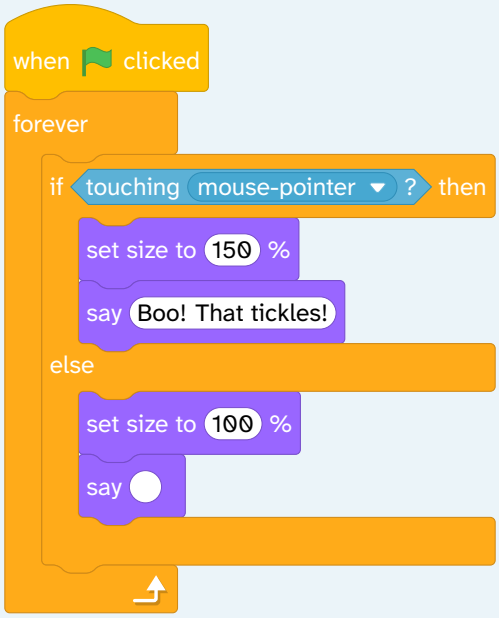
1. New project. From **Events**, drag four "when space key pressed" blocks. Click each dropdown and choose **right arrow**, **left arrow**, **up arrow** and **down arrow**.
2. Under each, snap a **Motion** block: "change x by 10" (right), "change x by -10" (left), "change y by 10" (up), "change y by -10" (down).
3. Click the stage once, then press the arrow keys. Save as `explorer`: you need it for 3.3!



x is left and right. **y** is up and down. Which key makes y bigger? _____

Exercise 3.2: Tickle the Ghost

1. New project. Delete the cat (right-click → **delete**). **Choose a Sprite**: pick **Ghost**.
2. Build the script. The pointy "touching mouse-pointer?" block is in **Sensing**. For the last "say", delete the text so the box is empty.
3. Click the flag, then move your mouse over the ghost and away again.

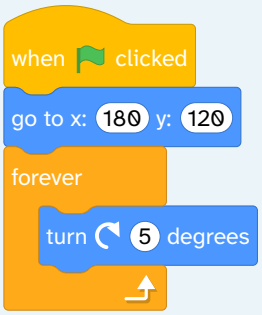


Predict first: what will happen if you take away the “forever”?

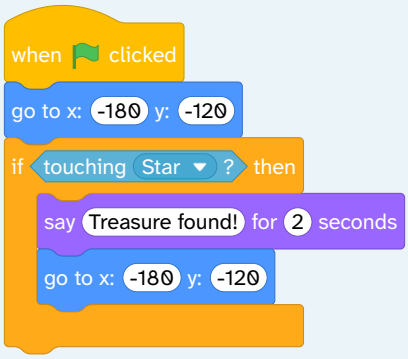
Exercise 3.3: Treasure Hunt

Steer your explorer to the treasure!

1. Open your `explorer` project (**File** → **Load from your computer**).
2. **Choose a Sprite**: add a **Star**. Give the star this script so it waits in the top-right corner and spins:



Bug hunt! Sam gave the explorer this script:



3. Click your explorer sprite and build Sam’s script. Click the flag and steer onto the star. What happens?

4. Why? Fix it! (Remember the goalkeeper.)

Done when: every time the explorer touches the star, it says “Treasure found!” and goes back to the start.

Vocabulary box

event Something that happens (a click, a key press) that starts a script.

condition A yes/no question, like “touching edge?”.

if Runs the blocks inside only when the condition is yes.

if-else Runs one set of blocks for yes and another for no.

sensing Blocks that let a sprite notice things: touching, keys, the mouse.

feedback Comments that help someone improve their work.

Reflection: Two stars and a wish

Show your project to a partner. Write what they said.

★ Star 1:

★ Star 2:

Wish (an idea to make it better):

What is the difference between “when space key pressed” and “if touching…”?

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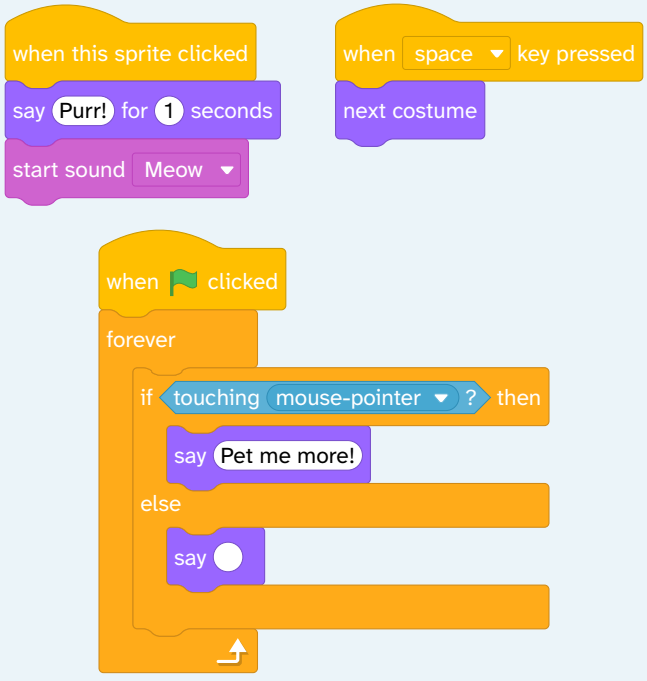
15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Pet Reactions (Scratch, about 10 minutes)

Make a virtual pet that reacts to you in three different ways.

1. Open Scratch. Delete the cat or keep it as your pet. Or click **Choose a Sprite** and pick any animal.
2. Build the three scripts below. Each one has a different doorbell (event).
3. Click the flag. Then: click the pet, press `space`, and move the mouse over the pet.
4. Save as `pet-reactions`.



(The sound block may have a different name for your animal: just pick any sound from the dropdown.)

★★ Challenge (want more?): Find the bug

Kai wants the pet to say “Wake up!” whenever the mouse touches it. Kai’s script is: “when green flag clicked” → “if touching mouse-pointer? then say Wake up!”. It never says “Wake up!”. Why not? How would you fix it?

Creative (make it yours): Two stars and a wish at home

Show your pet to someone at home. Ask them for two stars (two things they like) and a wish (one idea). Write them here, then try to make the wish come true!

★ _____

★ _____

Wish: _____

Self-check: tick each box when it is true

- ☐ My pet reacts to a click, a key press and the mouse.
- ☐ My “if” is inside a “forever”, so it keeps checking.
- ☐ I can say which block is the event (doorbell) in each script.
- ☐ When I gave or got feedback, it was kind and specific.